

Plant Doctor Handbook!



Created by the British Society for
Plant Pathology

Plants get sick too!

Just like us, plants get sick when microorganisms called pathogens attack them. Plant pathogens are very good at attacking plants and can cause massive amounts of damage.



Credit Tom Maack

Why do plant diseases matter?

Plants are essential to all aspects of human life. They provide us with food, oxygen, clothing, medicines and lots more! Without them we would not survive, and so it's vital we understand what's attacking them. And that's where plant doctors come in!

What attacks plants?

Fungi

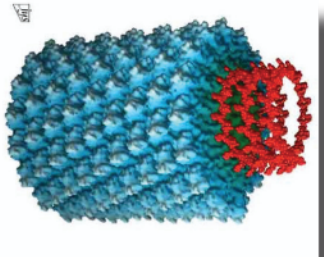
- Common plant attackers and can destroy even the largest trees!
- Often live in the soil where they attack plant roots.
- Absorb water and nutrients from plant roots.
- Release spores, which help fungi spread to new plants.



Credit J.Marqua

Viruses

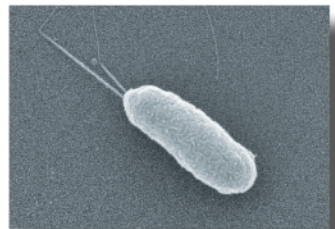
- Injected into plant tissue by insects like aphids.
- Turn plants into a virus making machines!
- Reduce plant growth, harming or even killing the plant.



Credit Grasp image by J.Y. Sgro, UW-Madison, from PDB

Bacteria

- Enter plants through wounds on the plant surface.
- Often have long protein tails called flagella that help them swim through the soil towards plants.
- Once inside the plant they kill off plant tissue using toxins, giving the bacteria an area in which to reproduce.



Credit Dr Robert Jackson, BSPP

Plant Doctor Training

Step 1

Which human diseases can you name? What do you think causes these diseases?

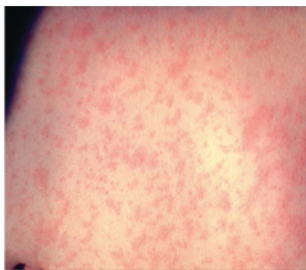
Can you match these human diseases with plant diseases?
Why do you think the diseases look similar?



Credit PD-USGov-HHS-CDC



Credit BSPP



Credit PD-USGov-HHS-CDC



Credit Pollinator, Wikimedia



Credit James Heilman MD,
Wikimedia



Credit Florida Division of Plant
Industry Archive, Bugwood.org

Step 2

What are the symptoms? Do you think this fruit has a fungal, bacterial or viral disease?

Symptoms:

Step 3

Can you draw a picture to show the differences between a healthy plant and a sick plant?

Healthy plant	Diseased plant

Where are they?

In our history: Irish Potato Famine

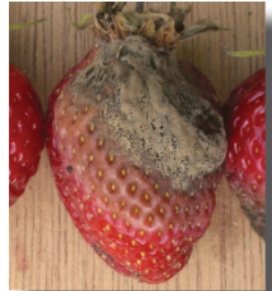
Plant diseases have devastating effects on food crops and even lead to famine. The Irish potato famine happened in 1845 when an outbreak of late potato blight disease wiped out Ireland's potato crop. Of the Irish population (8 million) approximately 1 million people died of starvation and 1.5 million emigrated.



Credit Rasbak

In our gardens and fruit bowls

You will probably have seen plant diseases in gardens and parks. Common diseases include rose black spot, which is caused by a fungus. If you leave fruit for too long you'll notice it grows a grey fluffy mould. This is caused by the *Botrytis cinerea* fungus. The fluffiness is due to the production of thousands of tiny spores, which help the fungus spread to new plants.



Credit Aardbei Lambada vruchtrot

In our woodlands

As of 2012 the ash dieback fungal disease is threatening our ash trees. In the 1960's an outbreak of Dutch elm disease (also fungal) devastated UK elm trees. It has since killed more than 25 million trees in the UK alone. Now very few elms are left in the UK or Europe.



Credit Luis Fernández García L. Fdez

Did you know?

The heaviest living thing on earth is... the honey fungus

One honey fungus in Oregon, USA has colonized 3.4 square miles of woodland. That's around the length of 88 football pitches! It has taken around 2,300 years to become this big and in that time it has killed a lot of trees!



Credit Stu Phillips

Honey fungus mushrooms can actually glow in the dark! It may help attract animals that can spread their reproductive spores. Or it may attract insects that eat smaller insects feeding on the fungus.

Trees get snot too!

Just like you, trees can catch a cold. A strain of the *Pseudomonas syringae* bacteria infects horse chestnut trees, causing bleeding canker disease. Here the bacteria multiply under the tree bark, causing the tree to ooze sticky snot from its trunk.



Credit F lamiot

Call the Plant Doctors!

A group of scientists called plant doctors or plant pathologists research plant diseases. They investigate plant pathogens and what makes them good at attacking plants. They also study how plants respond to pathogens and why some plants are better able to resist pathogen attack. Understanding this will help us grow plants that are more resistant to attack, saving our money, food, plants and trees!



Future Plant Doctors

The British Society for Plant Pathology is a large international group of scientists committed to understanding plant pathology. Part of their work includes organising outreach events to help people understand the importance of plant diseases and hopefully inspire future plant doctors!



Credit BSPP